

# From 5 Million Dirty Rows to One Trusted Catalog

A consumer-packaged-goods data story — 5,014,409 catalog rows modelled on Libstar's public brand portfolio, cleaned by Azure Data Factory, served by Synapse serverless SQL, scored with machine learning, and reconciled to the rand across Qlik Sense, Excel and this page.

**Case-study note:** Libstar is a real JSE-listed CPG group, but every record here is synthetic — generated (seeded, reproducible) on Libstar's *public* brand/category model to demonstrate the pipeline without using any private data. Not affiliated with Libstar Holdings.

**462 825**

raw rows generated (incl. planted duplicates)

**388 504**

clean unique products after ADF

**66 136**

rows quarantined — each with a reason

**R1.17tn**

trailing 12-month revenue (synthetic)

0.71

price-model  $R^2$  · MAE R12.71

## 1 · More than a product list

CPG catalogs rot quietly. Prices arrive as "R 1,234.56" from one system and "ZAR1234" from another; weights get captured in grams instead of kilograms; brands are typed by hand; category codes from a previous org structure ("Groceries", "Snacks and confectionery") linger years after a restructure. None of it looks urgent — until mispriced SKUs erode margin, duplicate listings distort demand forecasts, and rollup reporting breaks exactly when a range review needs it.

This project demonstrates the governed fix: a pipeline that **quantifies** the mess instead of silently filtering it, resolves it with auditable rules, and publishes one set of numbers that every downstream consumer — the live Qlik Sense app, the Excel pack, the ML models, this page — reconciles to **exactly**.

## 2 · The pipeline: dirty in, trusted out

```
generate_dirty_products.py (5.1M rows, seeded mess)
  | gzip CSV
  ▼
ADLS Gen2 raw/ → reference/ (brand map + dims)
  |
  ▼
Azure Data Factory · mapping data flow (8 vCores, ~6 min, ≈ R10/run)
  price/date/unit/boolean normalisation · brand lookup resolves misspelled
  brands AND legacy category names · dedupe (latest wins) · quarantine
  | snappy parquet
  ▼
```

ADLS Gen2 curated/ → Synapse serverless SQL (rpt.\* views, \$0 idle)

- Qlik Sense Cloud app (5 sheets, built via qlik-cli)
- scikit-learn (pricing · IsolationForest · KMeans)
- Excel report (formula-driven, reconciled)
- Ebook + this data story

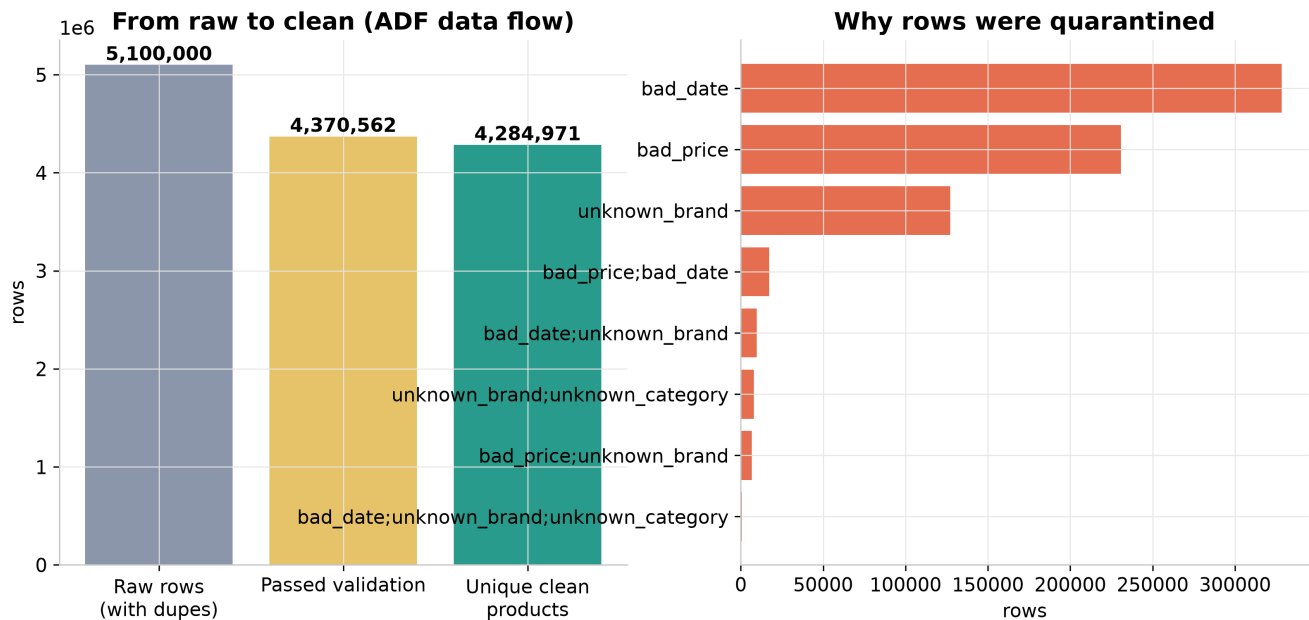


Figure 1 — 5,014,409 deduplicated rows in, 4,284,971 unique clean products out. Dates and prices are the biggest offenders.

### 3 · What the shelf looks like

Libstar's simplified operating model has two product groups. Ambient categories carry **77% of revenue** (R904.1bn of R1,170.3bn), led by Spreads, Meal ingredients and Wet condiments; perishables — dairy, convenience meals, value-added meats, baby, fresh mushrooms — hold the rest. Margins cluster tightly around the 35% portfolio average, so revenue mix, not margin spread, is what moves the total.

Revenue by category — Ambient does the heavy lifting

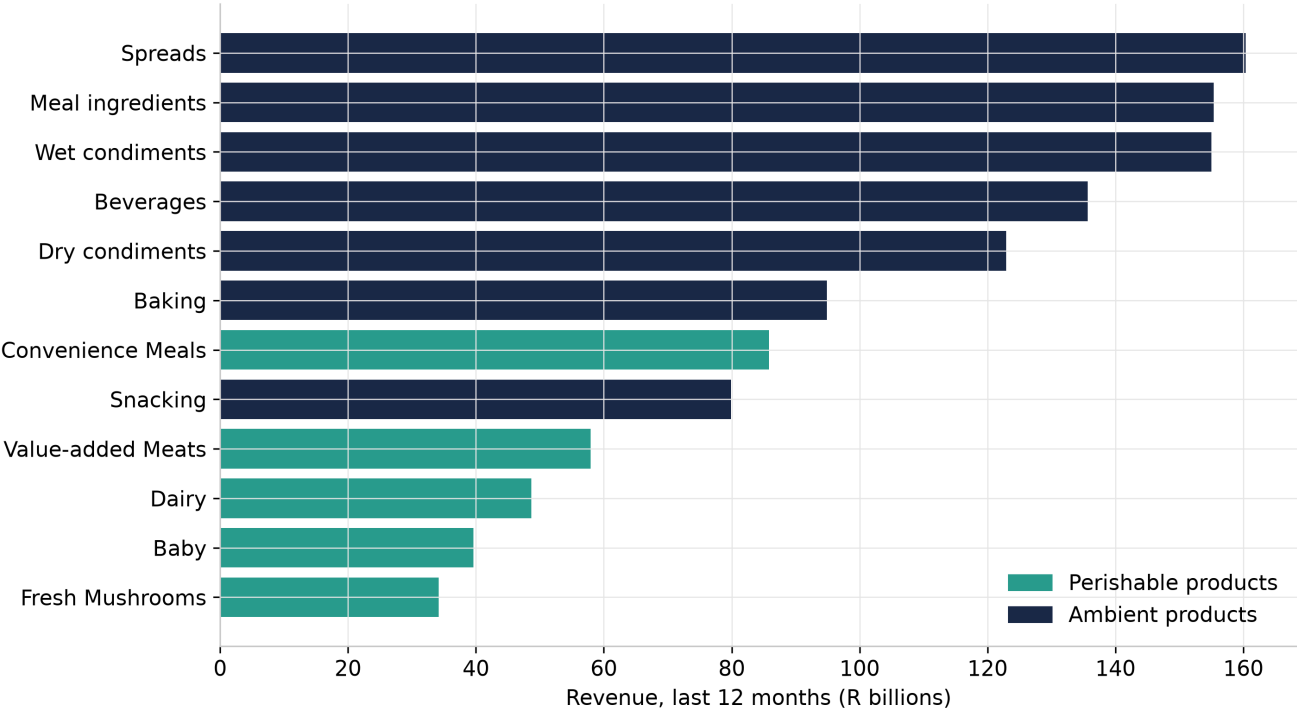


Figure 2 — Revenue by category, coloured by product group.

Margin by category

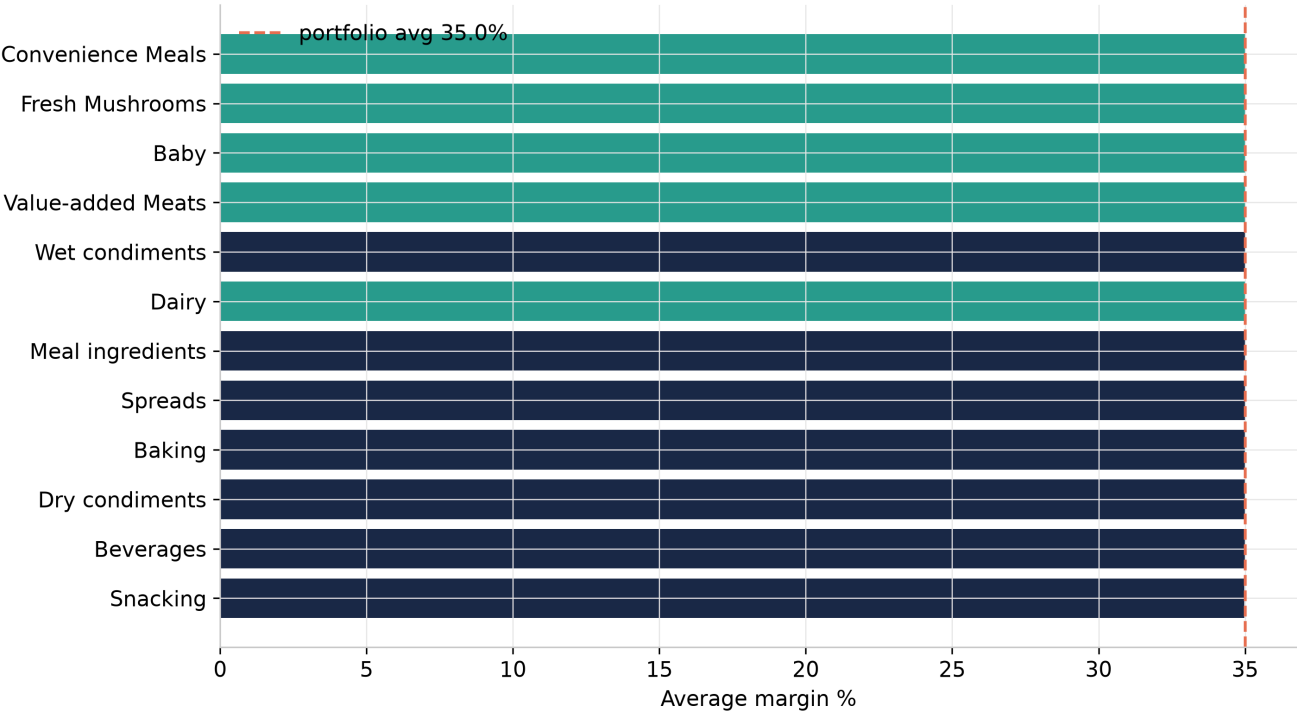


Figure 3 — Margin by category vs the portfolio average.

## Category scorecard (full population)

| Category          | Group               | SKUs    | Revenue 12m      | Avg margin | Avg price |
|-------------------|---------------------|---------|------------------|------------|-----------|
| Spreads           | Ambient products    | 526,813 | R160,395,514,140 | 35.0%      | R87.63    |
| Meal ingredients  | Ambient products    | 554,781 | R155,347,925,715 | 35.0%      | R74.20    |
| Wet condiments    | Ambient products    | 580,617 | R154,977,096,220 | 35.0%      | R68.20    |
| Beverages         | Ambient products    | 428,482 | R135,697,476,592 | 35.0%      | R94.98    |
| Dry condiments    | Ambient products    | 452,143 | R122,903,791,560 | 35.0%      | R69.79    |
| Baking            | Ambient products    | 427,832 | R94,861,638,385  | 35.0%      | R46.87    |
| Convenience Meals | Perishable products | 295,388 | R85,861,274,105  | 35.0%      | R79.71    |
| Snacking          | Ambient products    | 311,271 | R79,899,697,159  | 35.0%      | R62.37    |
| Value-added Meats | Perishable products | 176,523 | R57,959,630,790  | 35.0%      | R102.24   |
| Dairy             | Perishable products | 194,459 | R48,642,328,237  | 35.0%      | R59.31    |
| Baby              | Perishable products | 168,555 | R39,589,869,572  | 35.0%      | R52.17    |
| Fresh Mushrooms   | Perishable products | 168,107 | R34,146,064,405  | 35.0%      | R39.17    |

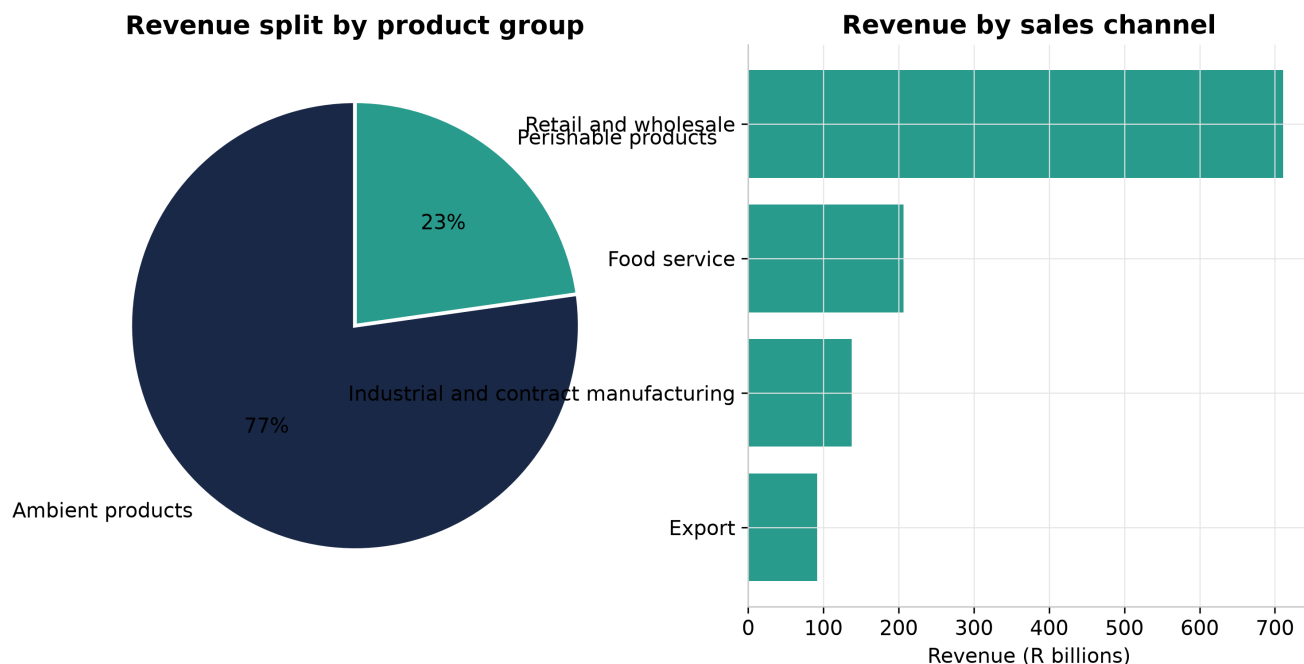


Figure 4 — Product-group split and the four routes to market.

## 4 • Brand power: three ways to win

Three brand solutions compete on the same shelf: Libstar's own brands (Lancewood, Denny Mushrooms, Cape Herb & Spice...), Principal brands represented in South Africa (Bonne Maman, Kikkoman, Tabasco, Maille), and private label for retailers. Principal brands price ~35% above comparable own-brand products; private label sits ~20% below — yet "Retailer Brands" is the single biggest revenue line (R103.7bn) because it spans all twelve categories.

### Top 15 brands by revenue

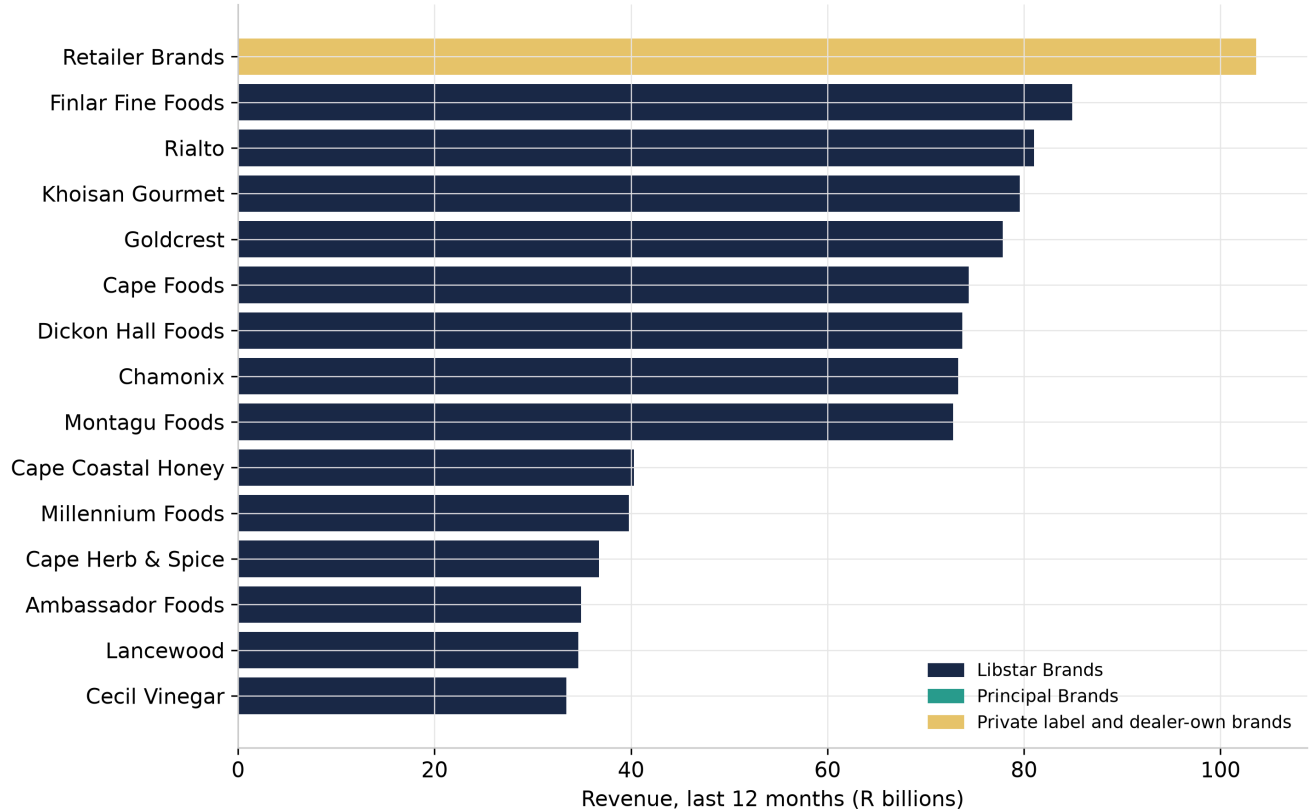


Figure 5 — Top 15 brands by revenue, coloured by brand solution.

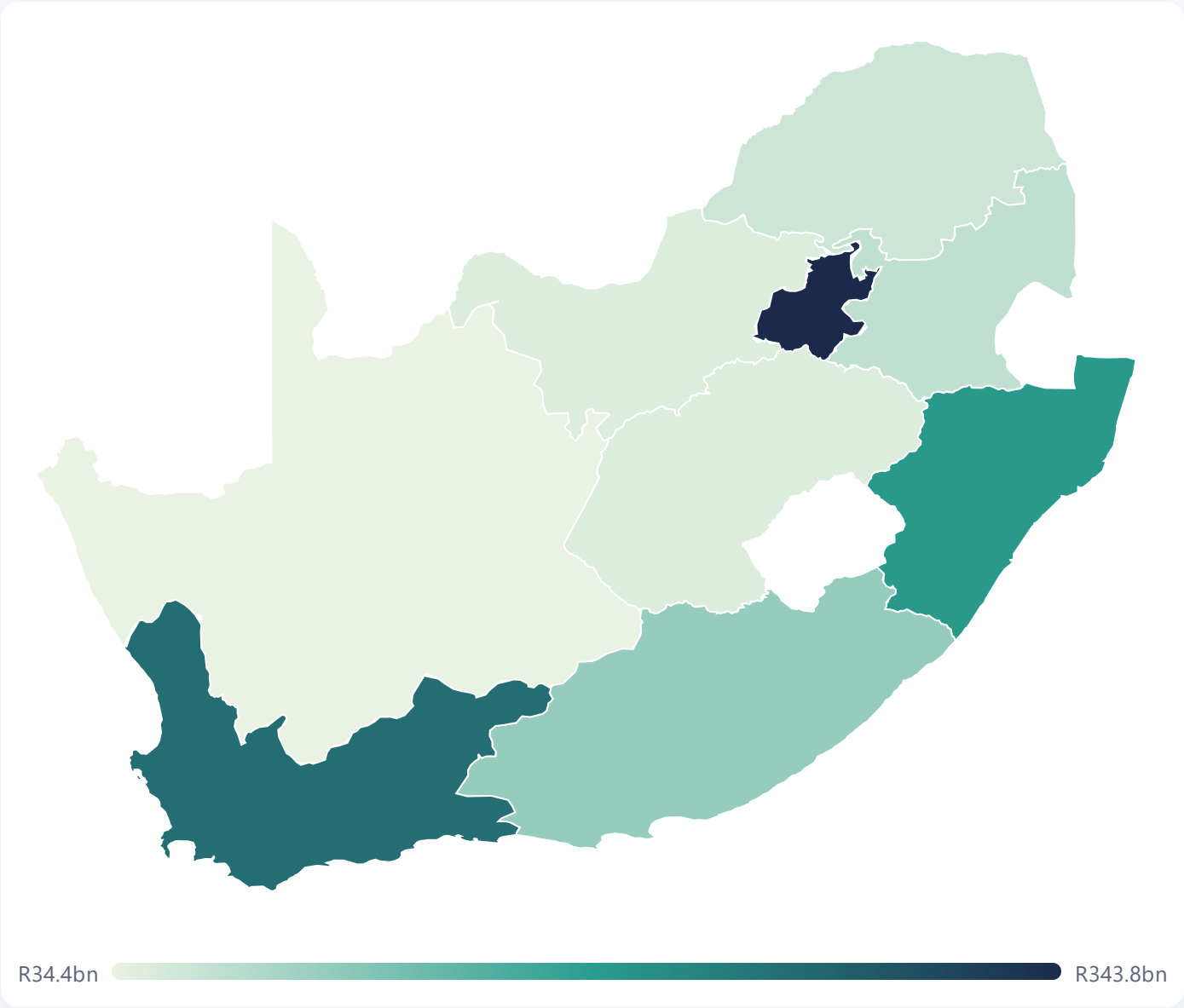
### Top 10 brands

| Brand             | Brand solution                      | SKUs    | Revenue 12m      | Avg margin |
|-------------------|-------------------------------------|---------|------------------|------------|
| Retailer Brands   | Private label and dealer-own brands | 430,236 | R103,650,387,104 | 35.0%      |
| Finlar Fine Foods | Libstar Brands                      | 268,984 | R84,928,535,830  | 35.0%      |
| Rialto            | Libstar Brands                      | 269,175 | R81,006,558,210  | 35.0%      |
| Khoisan Gourmet   | Libstar Brands                      | 267,412 | R79,563,401,192  | 35.0%      |
| Goldcrest         | Libstar Brands                      | 268,612 | R77,820,300,503  | 35.0%      |
| Cape Foods        | Libstar Brands                      | 268,381 | R74,364,367,070  | 35.0%      |
| Dickon Hall Foods | Libstar Brands                      | 269,141 | R73,699,851,203  | 35.0%      |
| Chamonix          | Libstar Brands                      | 269,004 | R73,286,135,131  | 35.0%      |

|                    |                |         |                 |       |
|--------------------|----------------|---------|-----------------|-------|
| Montagu Foods      | Libstar Brands | 269,449 | R72,784,041,446 | 35.0% |
| Cape Coastal Honey | Libstar Brands | 134,815 | R40,291,885,412 | 35.0% |

## 5 • Where South Africa buys

Hover (or tap) a province. Gauteng leads (R343.8bn), then Western Cape and KwaZulu-Natal — sales concentrate in the five provinces where the business has production or distribution footprints.





## Hover a province

12-month revenue



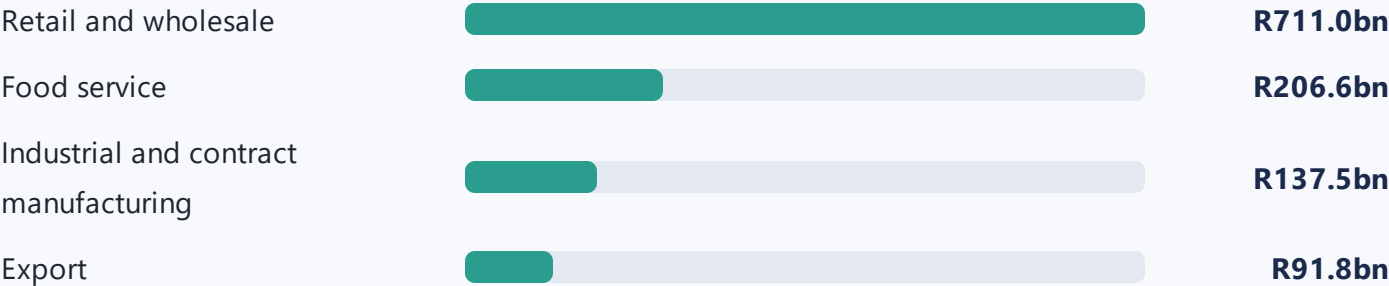
Products stocked



Average margin



## Routes to market



# 6 · Data quality — where the numbers leak

The Data Factory flow rejected **729,438 rows (14.5%)** — and can say exactly why, row by row. Because reject reasons ship into the same reporting layer as the clean data, data quality becomes a trend a team can manage down, not an invisible loss.

| Reject reason        | Rows    |
|----------------------|---------|
| bad_date             | 328,279 |
| bad_price            | 230,750 |
| unknown_brand        | 127,137 |
| bad_price + bad_date | 17,200  |

|                                                         |       |
|---------------------------------------------------------|-------|
| bad_date + unknown_brand                                | 9,759 |
| unknown_brand + unknown_category                        | 8,039 |
| bad_price + unknown_brand                               | 6,615 |
| bad_date + unknown_brand + unknown_category             | 657   |
| bad_price + bad_date + unknown_brand                    | 516   |
| bad_price + unknown_brand + unknown_category            | 450   |
| bad_price + bad_date + unknown_brand + unknown_category | 36    |

**Reconciliation guarantee:** the Qlik app, the Excel report, the ebook and this page all read the same aggregate layer. Revenue is **R1,170,282,306,879** in all four — to the rand.

## 7 • Machine learning: pricing & anomalies

**Pricing has learnable structure.** A gradient-boosted model predicts shelf price from category, brand solution, channel, weight and rating with  $R^2 = 0.71$  and a mean absolute error of **R12.71** on a mean price of R71.69 — evidence that pricing follows the portfolio's architecture, exactly what a well-governed catalog should show.

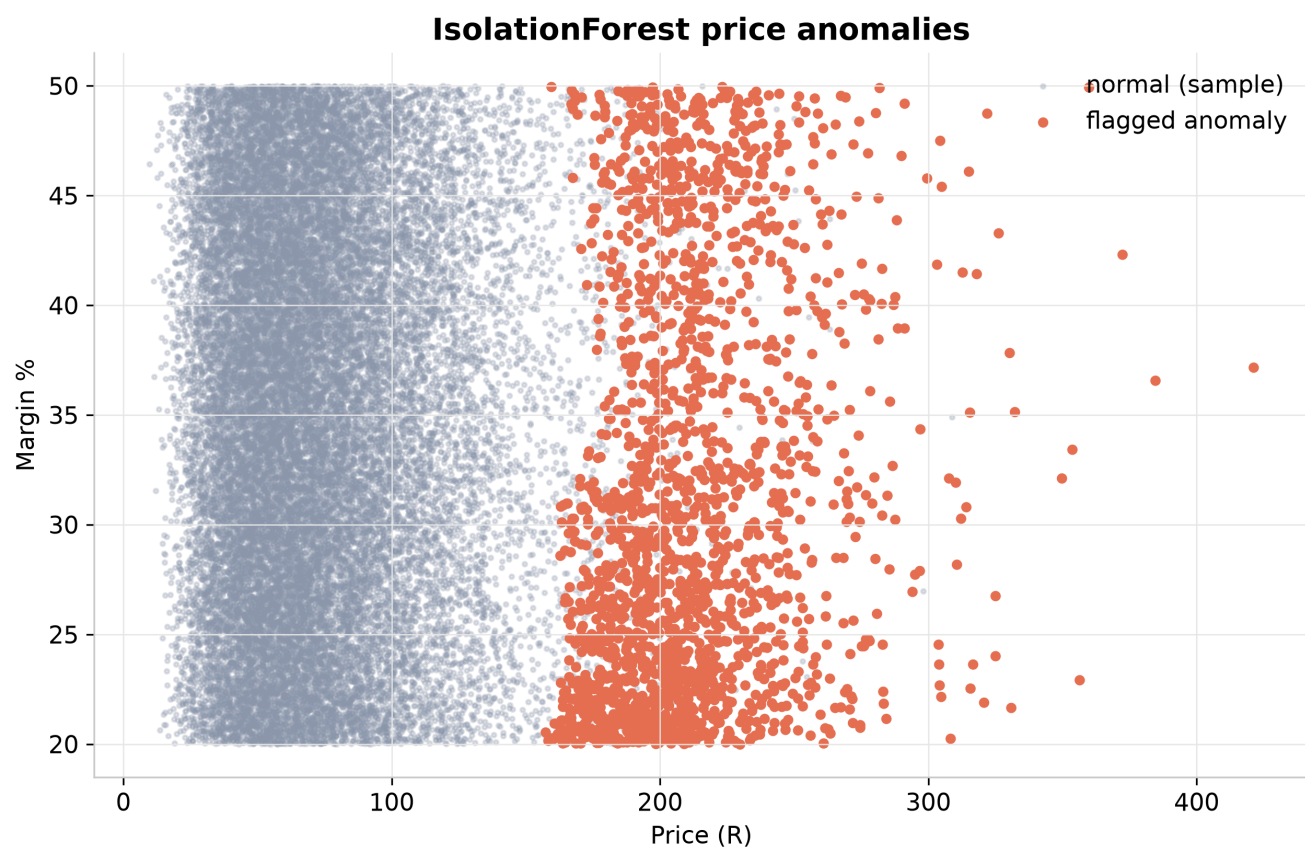


Figure 6 — IsolationForest anomalies vs the normal price–margin cloud.

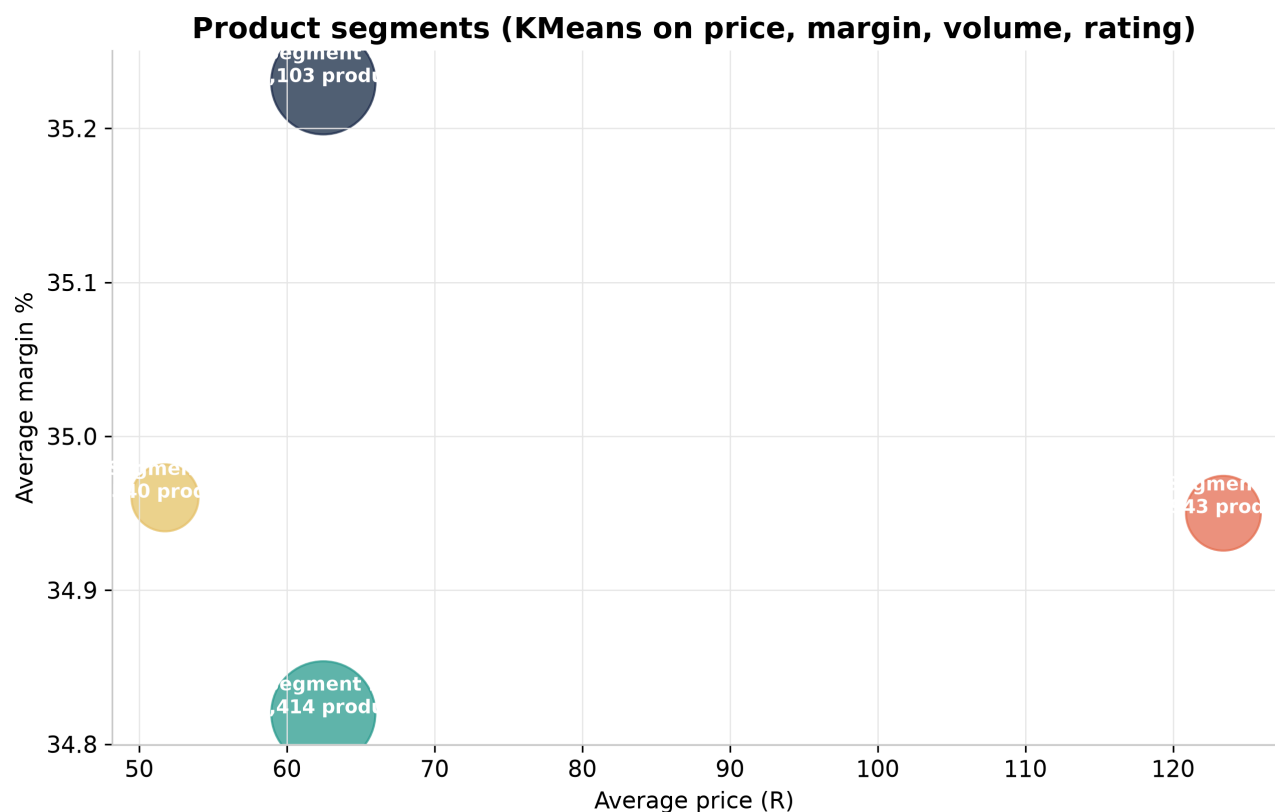


Figure 7 — Four KMeans segments: volume movers, premium niche, two mainstream clusters.

**Why items get flagged, in plain language:** the model scored one million products on price, cost, margin and weight *together*, and flagged the 2,000 whose combination is most unlike the rest — a heavyweight item priced like a small pack, an implausible margin, a luxury-priced commodity. These usually signal data-entry or pricing errors, not genuine premium products. **Recommendation:** route this list to category managers before the next price-list publication — correcting them before they reach shelves protects margin and avoids customer-facing pricing errors.

## Top 10 to investigate first

| Product                           | Category          | Brand             | Price   | Cost    | Margin | Score   |
|-----------------------------------|-------------------|-------------------|---------|---------|--------|---------|
| Classic Beef Burger Patties 250ml | Value-added Meats | Finlar Fine Foods | R227.69 | R181.82 | 20.1%  | -0.0801 |

|                                   |                   |                    |         |         |       |         |
|-----------------------------------|-------------------|--------------------|---------|---------|-------|---------|
| Premium Orange Marmalade 750g     | Spreads           | Bonne Maman        | R260.87 | R208.63 | 20.0% | -0.0791 |
| Everyday Hot Chocolate 250ml      | Beverages         | Rialto             | R315.01 | R169.81 | 46.1% | -0.0760 |
| Classic Strawberry Preserve 2kg   | Spreads           | Dickon Hall Foods  | R193.94 | R154.88 | 20.1% | -0.0749 |
| Select Chicken Nuggets 250g       | Value-added Meats | Finlar Fine Foods  | R210.95 | R168.58 | 20.1% | -0.0749 |
| Everyday Chicken Nuggets 500ml    | Value-added Meats | Finlar Fine Foods  | R224.52 | R179.44 | 20.1% | -0.0749 |
| Traditional Mac & Cheese Meal 1kg | Convenience Meals | Millennium Foods   | R219.97 | R174.70 | 20.6% | -0.0748 |
| Classic Instant Cappuccino 250ml  | Beverages         | Chamonix           | R281.27 | R155.00 | 44.9% | -0.0739 |
| Organic Green Tea 500ml           | Beverages         | Chamonix           | R203.88 | R162.23 | 20.4% | -0.0732 |
| Gourmet Apricot Jam 2kg           | Spreads           | Cape Coastal Honey | R236.00 | R186.40 | 21.0% | -0.0724 |

## 8 · Azure in production

Everything runs on a free-trial credit by design — no always-on compute anywhere in the chain.

| Resource             | Purpose                           | Idle cost             |
|----------------------|-----------------------------------|-----------------------|
| stpargoparcelis01    | ADLS Gen2 — raw / curated zones   | ≈ \$0.03/month        |
| adf-pargoparcelis-za | Data Factory — cleaning data flow | \$0 (per-run billing) |

`syn-pargoparcels`

Synapse serverless SQL endpoint

\$0 (\$5/TB scanned)

`pargo-trial-budget`

\$200 budget · email alerts at 50/75/90%

free

Kill-switch once all consumers are loaded: `az group delete --name rg-pargoparcels --yes`. The Qlik app (data loads in-app), the 36 MB .qvf export, the Excel pack, the ebook and this page all keep working after teardown.

## 9 · How it's built

| Layer            | Tools                                                                             |
|------------------|-----------------------------------------------------------------------------------|
| Data generation  | Python (pandas, numpy) — seeded synthetic catalog, reproducible mess              |
| Storage          | Azure Data Lake Storage Gen2 (raw / curated zones, Git LFS mirror in repo)        |
| Transformation   | Azure Data Factory mapping data flows                                             |
| Serving          | Azure Synapse Analytics serverless SQL                                            |
| Machine learning | scikit-learn — HistGradientBoosting, IsolationForest, KMeans                      |
| BI               | Qlik Sense Cloud — app, measures and 5 sheets built programmatically via qlik-cli |
| Reporting        | openpyxl (Excel) · python-docx (ebook) · matplotlib (charts + SA map) · this page |
| Ops & cost       | Azure CLI, PowerShell, Cost Management budget alerts                              |

Rebuild from a clean clone:

```
git lfs pull
python scripts/build_reference_dims.py && python scripts/build_brand_map.py
python scripts/generate_dirty_products.py          # ~15 min
adf\deploy_adf.ps1 → run pl_clean_products → synapse\deploy_synapse.ps1
python ml/train_models.py && python scripts/make_charts.py
```

```
python scripts/build_excel_report.py && python scripts/build_ebook.py  
python scripts/build_data_story.py # this page
```

Libstar Product Analytics — synthetic portfolio project by Anthony Apollis · July 2026

Data modelled on Libstar's public brand/category structure (libstar.co.za) · Not affiliated with Libstar Holdings